

# InnoColor Edge: A Real-Time Wearable System with Distilled Vision Transformer Guided Recoloring for Color Vision Deficiency

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## Abstract

Daltonization, or image recoloring for individuals with color vision deficiency (CVD), utilizes color clustering, gamut mapping, and more recently, learning-based computer vision methodologies to transform images for CVD perception. However, current methods face trade-offs between contrast enhancement, naturalness, and computational efficiency. To address these limitations, this study proposes a novel deep-learning framework that fully utilizes recoloring schemes learned by a Swin transformer to train the Dense Distillation Perceptual Algorithm (DDPA). This is the first time that a distillation-based methodology is used for image recoloring for CVD, producing a robust algorithm to overcome prior limitations of low contrast enhancement, narrow applicability, high computational cost, and edge-end infeasibility. DDPA preserves color consistency, reduces parameter count, and enables real-time edge deployment, achieving a global chromatic diversity (GCD) of 0.506, total color contrast (TCC) of 0.887, average gradient norm (AGN) of 0.254; in terms of color consistency, a structural similarity (SSIM) of 0.87 is achieved. The algorithm recolors an image in 6.43 ms at 0.199 MACs, independent of resolution. To demonstrate the framework’s efficacy, a wearable device has been designed and fabricated by cost-efficient 3D printing, a TCS3472 ambient light sensor, and a bluetooth microcontroller. A user application then implements the recoloring framework onto the device, InnoColor Edge. Use of InnoColor Edge improves the distinguishability of natural scenes for daily activities of colorblind users in real-time, while serving as an environment for researchers and testers to capture image data for developing datasets. Through the use of the user data interface, CVD users can collect and edit image pairs to construct a dataset suited to individual needs for retraining personalized recoloring models. Finally, a human evaluation of InnoColor Edge is conducted, demonstrating the advantage of DDPA in perceived saliency, contrast, and naturalness, while maintaining similar results to that of the teacher model, corroborating the learning capability of the DDPA architecture. Integrating the algorithmic and hardware components of InnoColor Edge, significantly extends the applications of the Daltonization field. The distillation and dataset pipeline enables broad generalizability, allowing InnoColor Edge to assist users across all major CVD types, addressing the 95% of cases underrepresented in current research.

# 1 Introduction

With the rapid development of digital media and its role in daily activities, efficient and accessible delivery of visual information has become crucial for accessibility. A major component of visual communication is consistent color perception; however, individuals afflicted with color vision deficiency (CVD) perceive color differently from how they were originally designed, often leading to confusion and discomfort [1]. Colorblindness affects 300 million people in recent years [2,3], and although its mechanisms are well understood, no cure currently exists. Because visual communication is integral to modern life, CVD can cause long-term anxiety, uncertainty, and even limit academic and career opportunities.

CVD is classified by the affected cone photoreceptor’s type and severity of defect; protan, deutan, and tritan forms correspond to abnormalities (anomalous trichromacy) or absences (dichromacy) of long, medium, and short wavelength cones, respectively, with the tritan conditions being much rarer than the protan or deutan [3]. While 5% of CVD individuals are red colorblind (protanopic), most cases are classified under partial colorblindness (anomalous trichromacy) [2]. Colorblindness reduces the perceivable color gamut, impairing the ability to distinguish hues otherwise easily separable by color normal observers [4]. As shown in Fig. 1, individuals with protanopia perceive green as yellow and pink as blue. Various approaches have been proposed to align digital media with the

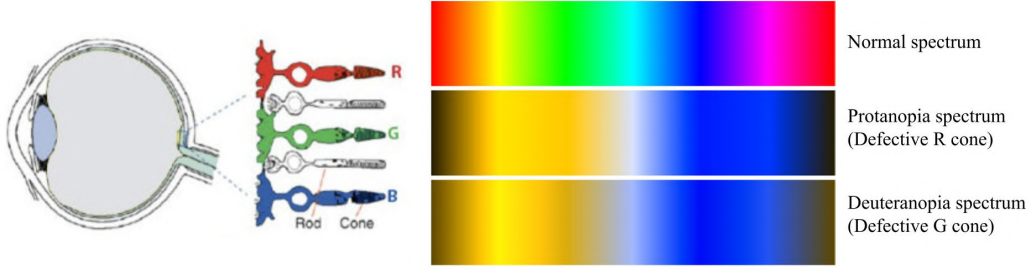


Figure 1: View of color spectrum after protanopia and deuteranopia conversion. Eye visual provided by <https://livanes.com.au/colour-blind-glasses-australia/>. Spectrum by researcher with Matplotlib, 2025.

perception of colorblind individuals, among which image recoloring, or Daltonization, has emerged as a reliable software based solution that adapts displays without requiring medical treatments [5]. Previous studies have also explored extending Daltonization algorithms to wearable devices [6]. With advances in deep-learning and computer vision, modern approaches now train models to enhance the contextual information in the image for the CVD viewer [7–9]. However, these high-performance models often remain impractical for day-to-day use due to their computational demands that cause a lack of compatibility with edge-end devices.

## 1.1 Objective of This Study

The objective of this study includes the following: 1) develop a distillation framework that trains the student model, the Dense Distillation Perceptual Algorithm (DDPA), to significantly improve the contrast enhancement, image naturalness, and computational compared to current recoloring al-

gorithms, 2) design and implement the InnoColor Edge system with a wearable device that provides daily visual enhancement for colorblind individuals, and 3) create a mobile and desktop application that integrates the framework into an augmented-reality interface for use within the wearable device, while enabling a user-based data generation pipeline to adapt InnoColor Edge to the needs of each CVD user.

## 2 Related Works

### 2.1 Summary of Previous Research

Few methods effectively and ecologically assist colorblind individuals, either in digital or physical settings. Through clinical testing, colorblind glasses such as Enchroma found no significant improvement in color discrimination [10]. Because these glasses rely on a single optical filter to address many color scenarios, their overall effect is minor [11]. Tanuwidjaja et al. instead proposed using Google Glass to augment physical perception through color highlighting or matrix-based recoloring, demonstrating potential improvements across different types of CVD [6]. While color highlighting can help identify specific shades, it depends on users knowing which color to target. This is an unrealistic assumption for daily use. Recoloring better addresses the general challenge of distinguishing multiple color regions, though the existing matrix-based methods simplifies psychophysical qualities of color perception and is not sufficient for the vast amount of use cases.

Comprehensive software recoloring algorithms can be categorized into numerical methods (hue-rotation [12], color-clustering [13, 14], palette recoloring [15], and objective optimization [16]) and learning-based methods. Numerical methods rely on rule-based measurements and color-space operations, offering lightweight computation and real-time contrast enhancement feasible for edge-end computing, though they cannot adapt to scene context. By contrast, learning-based methods have introduced computer vision algorithms. They extract contextual features such as regional color tones, object boundaries, and object identities using convolutional architectures [17–19]. Due to limited colorblind-specific data, most models employ general image classification datasets and train models through a CVD simulation filter [17, 18]. However, this does not resolve the psychophysical loss of color dimensionality that colorblind vision faces. Thus, Li et al. developed a paired image dataset specifically for protanopia recoloring, and proposed the use of image-to-image generative adversarial networks (GANs) to transform images while avoiding artifacts or unnatural colors [8]. Despite its advantages, the development of paired databases only for protanopia poses an issue of generalization. Adaptation to any different case of CVD is costly, requiring not only collection of a new dataset but also a redesign of adversarial network hyperparameters. Finally, compared to numerical algorithms, learning-based methods may prove more computationally demanding and difficult to implement in a daily setting.

Other learning-based approaches prioritize the application and deployment aspect of CVD recoloring, targeting computational feasibility as the main concern. To address the issue of time and memory costs, studies have opted for a lightweight method. I trained a set of learnable RGB

lookup tables and then used a CNN to predict the weighting of each lookup table for fusion, creating consistent recoloring under constant computational complexity [20]. This significantly reduces the degrees of freedom in image-to-image translation and ensures that similar colors in the original image appear similarly in the output. However, this method exhibits a disadvantage in regional contrast enhancement compared to Li et. al’s method, as network utilization is not sufficient and lookup-table learning is not optimized during training.

## 2.2 Identification of Research Gap

Despite numerous approaches to CVD recoloring, current research remains focused on the algorithm development stage, with little progress towards deployable hardware. To the best of my knowledge, no prototype wearable device currently achieves high recoloring performance within the computational limits of edge devices.

Most recoloring algorithms adapt classical deep-learning architectures such as classifiers and generative networks without considering the psychophysical model of human perception. As a result, despite achieving high objective performance in metrics, recolored images often appear either unnatural or fail to enhance perceived contrast between color regions. Prior methods also overlook chromaticity separation within the trichromatic color data, limiting learning capacity in compact networks. For larger, unsupervised networks, there is a lack of utilization of model distillation or parameter pruning, making inference computationally demanding for high-resolution images that appear frequently in digital media, such as HD videos or rasterized images.

Furthermore, no existing digital applications utilize high-performance frameworks for CVD recoloring, or provide user-driven customization to adapt tools to individual visual conditions.

## 2.3 Contributions of This Study

This study creates the InnoColor Edge system with the following contributions:

1. A deep-learning framework based on model distillation is developed to train the Dense Distillation Perception Algorithm (DDPA). DDPA demonstrates enhanced contrast (GCD, TCC, AGN); color consistency (SSIM); and greater computational efficiency in both runtime and operation count. Because metrics do not fully reflect perceptual quality, a human evaluation assesses both subjective evaluation and system-level usage of InnoColor Edge.
2. To demonstrate a proof-of-concept, a wearable device, InnoColor Edge, is designed and implemented for both daily use by CVD individuals to improve real-world color vision perception and by researchers for data collection within the InnoColor Edge pipeline. InnoColor Edge is integrated with a TCS3472 RGB light sensor through a bluetooth microcontroller, which for the first time, helps to detect ambient light conditions that affect color processing.
3. A mobile and desktop application are developed to integrate the deep-learning framework within the wearable device and data collection pipeline. Its toolbar applies DDPA and other

user-chosen models within the augmented-reality interface, while a data-generation interface enables users to customize datasets and adapt InnoColor Edge to their own perceptual profiles. This addresses over 95% of CVD cases that are unaccounted for in current research.

### 3 Methodology

#### 3.1 System Level Design of InnoColor Edge

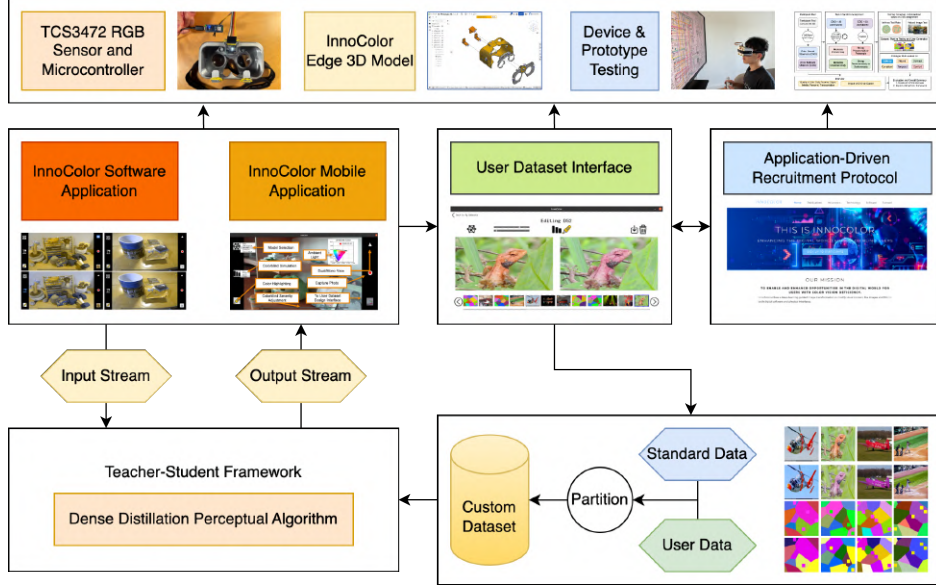


Figure 2: System level design of InnoColor Edge. Graphic created by researcher with draw.io, 2025.

As illustrated in Fig. 2, the InnoColor Edge system operates through a development feedback cycle. The main components include the wearable device, the native software applications, user data interface, data integration, and the deep-learning framework.

The distillation framework trains the novel deep-learning Dense Distillation Perceptual Algorithm (DDPA), using the student dataset generated by the teacher Swin transformer model, with future iterations incorporating user-collected data. DDPA generates a recoloring scheme based on the input image, and then applies it onto the input to produce the output. Its architecture is described in Section 3.2.1. The InnoColor Edge augmented-reality wearable device serves as a proof-of-concept for implementing an adaptive deep-learning framework and data pipeline. The device was fabricated for ease-of-use and control, and houses the TCS3472 RGB ambient light sensor during user interaction with the application. The InnoColor Edge wearable device is described in Section 3.3. Within the InnoColor Edge application, users view the live camera stream while using a toolbar to adjust CVD simulation parameters, recoloring models, and color highlighting, or to store images for data editing. A dataset editing and export interface further allows users to manually curate and contribute data to the InnoColor Edge pipeline. The user-generated data will support training and refinement of the deep-learning framework in future iterations. The user application

and data interface is described in Section 3.4.

### 3.2 Distillation Framework Development

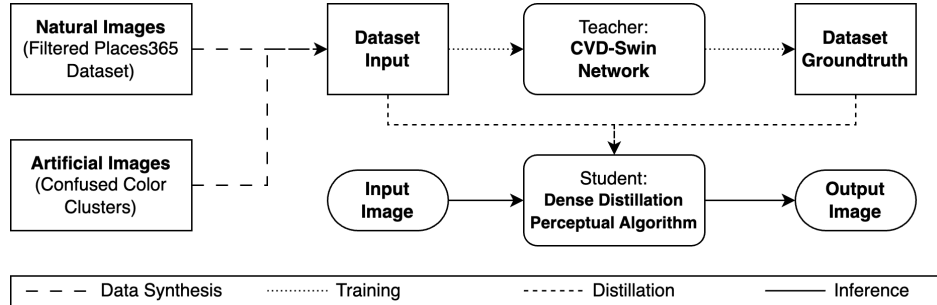


Figure 3: The proposed distillation framework, including data synthesis, training, and then inference. Graphic created by researcher with draw.io, 2025.

As demonstrated in Fig. 3, data synthesis is performed through the teacher model (CVD-Swin), with the dataset consisting of input images discussed in 3.2.2. The dataset inputs and outputs (groundtruth) are then used to train the DDPA, which is described in 3.2.1. Note that user data is not included for the initial framework development. The student algorithm DPPA demonstrates potential through high objective indicators and low hardware requirements for inference as discussed in Section 4. Currently, the framework is parametrized to recolor images for deuteranopic individuals (green colorblind). By modifying the colorblind simulation direction and severity parameters used by the CVD-Swin transformer to generate output images, the framework can be reoriented to recolor images for any form of CVD. Finally, the custom loss functions are described in Section 3.2.3.

#### 3.2.1 Development of Dense Distillation Perceptual Algorithm (DDPA)

Lookup-tables (LUTs) provides key advantages for recoloring that have not been utilized extensively in previous work. First, LUTs ensure color consistency, where two pixels of the same color in the input image will also contain the same color in the output image. Additionally, with the inclusion of smooth (total variance, TV [21]) loss discussed in 3.2.3, color consistency extends beyond identical hues, and ensures perceptual similarity among neighboring colors. As a result, similar colors in the input image will appear similar in the output image. Second, unlike prior methods that directly output a transformed image in entirety, DDPA processes a downsampled input to generate a LUT, significantly reducing redundant parameters in the convolutional network and preserving global color features.

As mentioned in Section 2, my previous research is the first to use LUTs, and uses RGB-based transformations [20]. However, its large parameter space ( $5 \times 17 \times 17 \times 17$ ) led to incomplete utilization of the LUTs and limited contrast enhancement.

The proposed method further constrains recoloring to chromaticity, reducing dimensions of mapping from three ( $RGB$ ) to two ( $uv$ ), helping to streamline model training and facilitating convergence.

Two  $L \times L$  LUTs encode  $u$  and  $v$  directional mappings, and are generated by a convolutional network  $NET$ . The input image  $I$  is first converted to  $yuv$ , downsampled to  $M \times M$ , and used to generate the LUTs. Next, the LUTs are applied to the  $u$  and  $v$  components of the original resolution image, and trilinear interpolation is used to retrieve values between lookup values. The  $y$  component is preserved and recombined. The  $yuv$  output is then converted to  $RGB$  color space, reconstructing the output image, recolored for CVD observers. The algorithm is represented below,

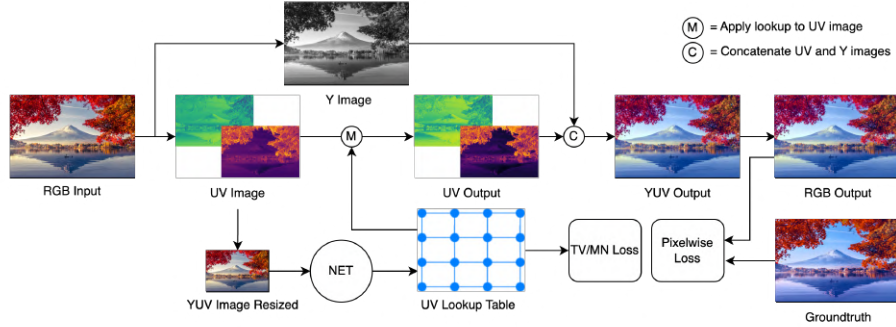


Figure 4: The methodology of DDPA is shown. Graphic created by researcher with draw.io, 2025.

$$O = \text{concat}(I_y, \text{LUT}_{uv}(I_{uv})) \quad (1)$$

where,

$$\text{LUT}_{uv} = \text{NET}(I_{uv, \text{down}}) \quad (2)$$

Importantly, by operating on a downsampled image, DDPA achieves constant computational complexity independent of image resolution, helping the InnoColor Edge system to scale efficiently to high-resolution images where recoloring is most often required.

### 3.2.2 Teacher Dataset Collection Process

The framework employs CVD-Swin as the teacher model, trained on an unpaired dataset with a selection of both natural images and artificially generated images to maximize regional color distinguishability in forward propagation [7]. Natural images were drawn from the Places365-Standard dataset containing around 1.8 million images [22] and filtered using the color contrast-preserving ratio (CCPR) to exclude scenes lacking color region combinations relevant to CVD perception [23], retrieving 41196 natural images. An additional 10000 synthetic images were generated by traversing RGB channels and assessing color confusion through  $Lab$  distance. Color sets were plotted as points on a canvas, and the rest of the canvas was filled out based on the nearest neighbor. To enhance generalization, the dataset was further augmented through adding rotational duplicates.

The use of an unpaired dataset confers a key advantage in generalizability. As the model is

not restricted to fixed input–output image pairs, it learns recoloring mappings through perceptual loss functions rather than groundtruth supervision. This allows the framework to incorporate any combination of CVD parameters, such as severity, type, or individual visual settings, as adjustable components of the loss function. During distillation, the flexible representations are learned by the student model DDPA, enabling it to generalize and recolor images for a wide range of users with differing CVD characteristics. The unsupervised teacher model within the distillation framework eliminates the need for compiling a new dataset for each CVD type, allowing InnoColor Edge to provide adaptive, personalized visual correction within a single unified system.

### 3.2.3 Calculation of Loss Function

DDPA is supervised by the  $L_1$  pixelwise reconstruction loss, total variation regularization, and monotonicity regularization.

DDPA adopts  $L_1$  for pixel-to-pixel reconstruction loss.

$$L_1 = \sum_{j=1}^h \sum_{i=1}^w |x_{i,j} - y_{i,j}| \quad (3)$$

where  $x, y$  denote the colors of homologous pixels in  $I, O$ , and  $h, w$  denote the image height and width.

In order to reduce structural discontinuities during LUT generation and maintain the smoothness of LUT transformation, TV and MN regularization terms are included [21]. The TV term minimizes color distance between adjacent LUT values, while the MN term ensures that the LUT color series is increasing, and does not change direction. The terms together help to control for undesirable artifacts shown in Section 4.5. First, the difference tables for the  $u$  and  $v$  dimensions are determined:

$$\mathcal{U}_{cuv} = \text{LUT}(c, u, v) - \text{LUT}(c, u + 1, v) \quad (4)$$

$$\mathcal{V}_{cuv} = \text{LUT}(c, u, v) - \text{LUT}(c, u, v + 1). \quad (5)$$

The losses are calculated correspondingly, with the TV term is minimized when the square difference between consecutive LUT values is reduced. The MN term is minimized when the LUT series is monotonically increasing, which reduces the ReLU terms to 0.

$$L_{TV} = \sum_{c,u,v} \mathcal{U}_{cuv}^2 + \mathcal{V}_{cuv}^2 \quad (6)$$

$$L_{MN} = \sum_{c,u,v} \text{ReLU}(\mathcal{U}_{cuv}) + \text{ReLU}(\mathcal{V}_{cuv}) \quad (7)$$

Regularization terms are scaled by coefficients  $\lambda_{TV} = 0.0001$  and  $\lambda_{MN} = 10$  respectively. The final loss equation for DDPA is listed below:

$$L_{DDPA} = L_1 + \lambda_{TV} L_{TV} + \lambda_{MN} L_{MN}. \quad (8)$$

### 3.2.4 Quantitative Evaluation Metrics

Seven metrics were selected for algorithm evaluation: local contrast (LC), global contrast (GC), total color contrast (TCC), global chromatic diversity (GCD) chromatic difference (CD), average gradient norm (AGN), and structural similarity (SSIM).

The comparative loss functions, LC and GC, are developed by Chen et al [7], and is based off the individual contrast loss. Given a pixel pair  $x, y$ , the contrast loss  $CL$  is defined as the difference between: (1) the color difference of the CVD-simulated pixel pairs in output  $O$ , or  $\hat{c}_x - \hat{c}_y$  and (2) the color difference of the original pixel pairs in input  $I$ , or  $c_x - c_y$ .

Since the CIELAB color space better aligns with human perceptual characteristics, this contrast loss is computed in CIELAB coordinates as the following:

$$CL(x, y) = \left| \left| \hat{c}_x - \hat{c}_y \right| - |c_x - c_y| \right| \quad (9)$$

$$\hat{c}_x = CVD(c'_x), \hat{c}_y = CVD(c'_y). \quad (10)$$

Here,  $c_x, c_y, c'_x, c'_y$  are the colors of the pixel pair  $x, y$  in images  $I, O$ , respectively.  $CVD(\cdot)$  denotes the CVD simulation for a color. Also,  $|\cdot|$  is the L1 norm for the pixel, which sums the color difference across the CIELAB channels.

The local contrast loss component computes pairwise distances between each pixel  $x$  in output  $O$  and pixels in its local neighborhood  $\omega_x$ . This term is formulated as:

$$LC = \sum_{x=1}^N \sum_{y \in \omega_x} \frac{CL(x, y)}{||\omega_x||}. \quad (11)$$

Here,  $||\omega_x||$  is the number of pixels in  $\omega_x$ .

Global contrast loss sums the individual differences for any randomly sampled pixel pair  $x, y$ . It is computed as follows:

$$GC = \frac{1}{||\omega||} \sum_{x, y \in \omega} CL(x, y). \quad (12)$$

Utilization of local and global contrast loss ensures alignment of CVD perception with normal perception for both specific regions of the image requiring contrast enhancement and the general color tone of the image.

Total color contrast (TCC) is introduced by Chen et al. [7]. It calculates the absolute contrast over all pixels within the image, relative to neighboring pixels:

$$TCC = \frac{1}{n_1} \sum_{(i,j) \in \Omega_1} (|x_i - x_j|) + \frac{1}{N * n_2} \sum_{i=1}^N \sum_{j \in \Omega_2} (|x_i - x_j|) \quad (13)$$

where  $\Omega_1$  is the set of points in the image,  $\Omega_2$  is the set of pixels surrounding a given point in  $\Omega_1$ ,  $n_1$  represents the pixel count in set  $\Omega_1$ , and  $n_2$  represents the pixel count in set  $\Omega_2$ .

The global chromatic diversity (GCD) value of an image measures the strength of color con-

trast, and is used by W. Chen et al., and Ma et al. for recoloring targeted towards dichromatic colorblindness [24, 25]. GCD is calculated as the sum of distances between each pixel pair:

$$GCD(i, j) = \sqrt{(l_i - l_j)^2 + (a_i - a_j)^2 + (b_i - b_j)^2} \quad (14)$$

where  $i$  and  $j$  lie in the color range of the CVD simulation of the recolored image, and  $l_i$ ,  $a_i$ , and  $b_i$  are  $L$ ,  $a$ , and  $b$  values of pixel  $i$ .

Chromatic difference (CD) is considered a naturalness metric, as it compares CIELAB values between  $I, O$ . It is summarized by Zhu and Mao [4]:

$$CD(i) = \sqrt{\lambda(l'_i - l_i)^2 + (a'_i - a_i)^2 + (b'_i - b_i)^2} \quad (15)$$

where  $l'_i, l_i, a'_i, a_i, b'_i, b_i$  are the CIELAB color values of the output and the input image, respectively, and  $\lambda$  is the coefficient of luminosity.

The average gradient norm (AGN) measures the local absolute contrast of the image. It is used by Z. Zhu et al [26]. This is calculated as

$$AGN(i) = \sqrt{G_h(i)^2 + G_v(i)^2} \quad (16)$$

where  $i$  ranges over the pixels in the CVD simulation result of the recolored image, and  $G_h(i)$  and  $G_v(i)$  represent the applications of the horizontal and vertical Sobel operator onto  $i$ .

To preserve the natural appearance of source images, the recolored output should faithfully approximate the original image. To quantify the perceptual similarity between recolored and original images, the SSIM metric is adopted. Note that while SSIM is deemed a naturalness metric, it does not encapsulate human-perceived naturalness, but rather serves as a mathematical objective for optimization.

$$SSIM(x, y) = \frac{(2u_x u_y + c_1)(2\sigma_{xy} + c_2)}{(u_x^2 + u_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \quad (17)$$

$$L_n = 1 - SSIM(O, I) \quad (18)$$

### 3.3 Design and Implementation of Device

The goal was to create a user-friendly device capable of implementing the recoloring framework (see Section 3.2) for both daily physical activities and digital activities such as entertainment, browsing, and research such as in Fig. 5. Design priorities included mobility, comfort, and cost-efficiency, given current hardware-software co-design limitations and the need for real-time image processing on low-power edge-end hardware.

To meet the design criteria, InnoColor Edge was developed as an augmented-reality wearable device: whereas other devices such as Google Glass or Meta Quest use fully-embedded hardware, InnoColor Edge is integrated with an iPhone as both display and processor. By leveraging existing

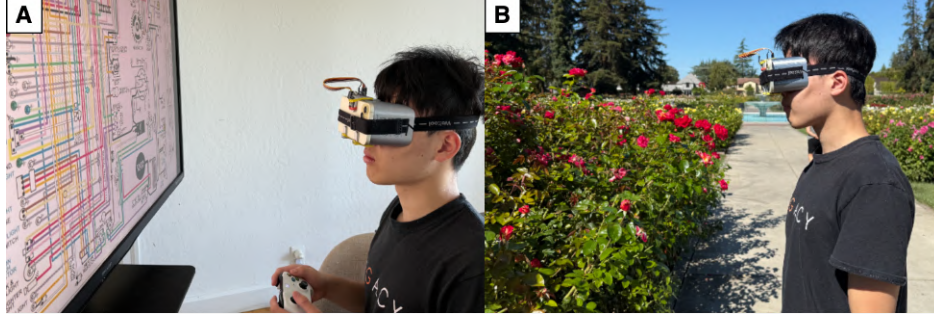


Figure 5: A) usage of InnoColor Edge in digital applications such as electric circuit diagrams. B) demonstration of device hands-free in natural scenery. Pictures captured by researcher, 2025.

hardware platforms, this method presents three benefits: first, stable software deployment across versions, facilitating future research; seconding, single integration of camera, display, and processor; and third, universal accessibility, eliminating additional costs compared with using components such as Raspberry Pi for computing capabilities. As detailed in Fig. 6, the 3D printed frame, modeled in

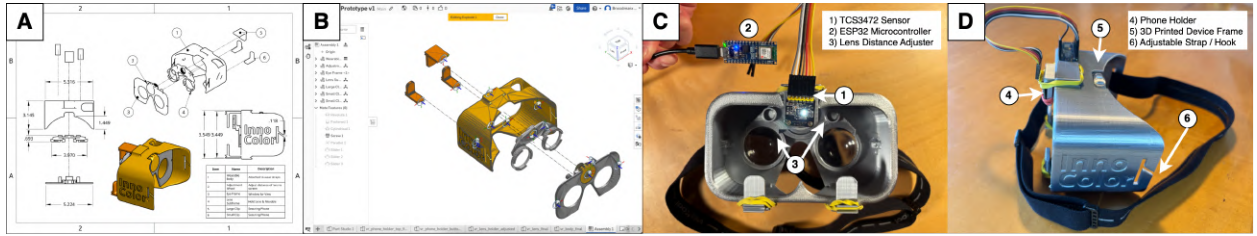


Figure 6: A, B) InnoColor Edge wearable device computer-aided design. C, D) Diagram of printed and assembled device. Graphics captured and created by researcher with Onshape, 2025.

Onshape, is designed to hold an iPhone and adjustable straps while maximizing comfort. Assembly confirms the freedom of movement. The augmented-reality lenses can be adjusted with a wheel, allowing users to modify the focal distance for visual comfort.

The design utilizes the ability to collect additional physical data. Importantly, applying a digital recoloring algorithm to raw and undoctored camera stream requires consideration of variances in lighting conditions in order to maintain color naturalness for the CVD user. To account for this, the device incorporates an ultra-high-sensitivity TCS3472 RGB ambient light sensor connected via I2C to an ESP32 microcontroller with a NORA-W106 bluetooth module. This configuration facilitates communication with both iPhone and desktop applications (the latter being used for testing outside the wearable device) while maintaining low energy consumption. The NORA-W106 module carries 8MB RAM, and the TCS3472 operates between 2.5-65  $\mu$ A.

As shown in Fig. 5, an Xbox controller can be also used to interact with the iOS application while wearing the device. While the Xbox controller is suited for developers and testers of the application within the InnoColor pipeline, it is not required for daily users to preserve comfort and practicality. The bluetooth protocol pairs to the iPhone directly, helping minimize interference to the microcontroller.

The implementation of the RGB ambient light sensor and Xbox bluetooth pairing within the

user application is described in Section 3.4.

As summarized in Table 1, the current prototype demonstrates a cost-effective design, with retail component prices totaling roughly \$30 USD. Under large-scale production, the cost can be reduced to around \$5.

Table 1: Price ranges for InnoColor Edge components. Created by researcher within Overleaf, 2025.

| Major Component                   | Min (USD)    | Max (USD)    | Current (USD) | Brand      |
|-----------------------------------|--------------|--------------|---------------|------------|
| TCS3472                           | 3.26         | 11.50        | 4.55          | Modulefans |
| ESP32 + NORA-W106                 | 18.30        | 30.76        | 21.00         | SparkFun   |
| Adjustable Elastic Strap          | 5.99         | 14.99        | 9.99          | Weltool    |
| Polylactic Acid Filament (\$/70g) | 0.70         | 2.10         | 1.40          | Bambu      |
| <b>Total</b>                      | <b>28.25</b> | <b>59.35</b> | <b>36.94</b>  |            |

### 3.4 Application

To fully enable deployment of the framework (see Section 3.2), a user application is developed on multiple hardware platforms. The application includes the augmented-reality interface with the InnoColor toolbar, and the user custom data generation interface. This is the first time that a comprehensive software application has been developed for visually aiding CVD users, providing a simulation and design platform for testers, and giving users the ability to generate and export their own data to the InnoColor Edge system.

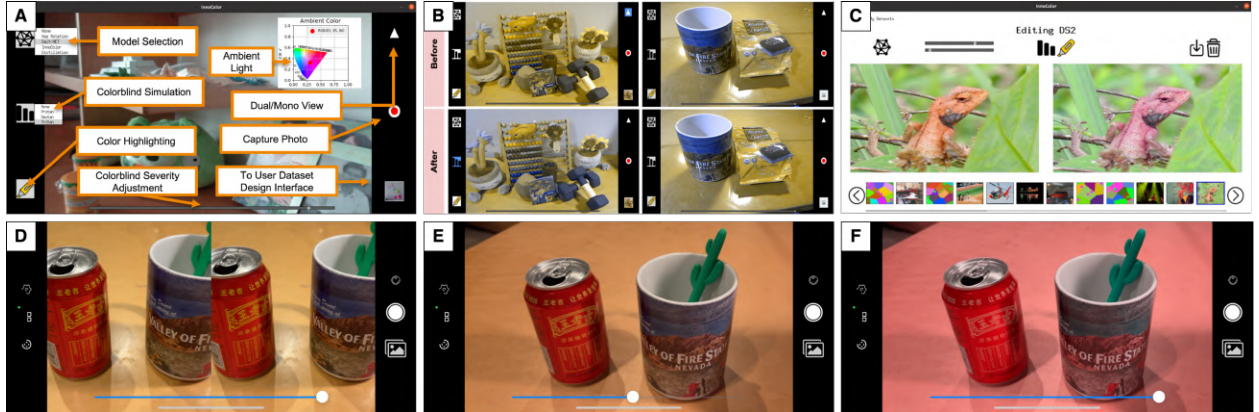


Figure 7: InnoColor Edge application. A-C) PC. D-F) Mobile interface. A) User Application Layout, including toolbar. B) Demonstration of DDPA. C) User-custom data generation interface, with carousel and toolbar. Images shown are from standard dataset. D) dual view when using wearable device. E) partial blue-blind simulation (shown by bottom slider). F) full blue-blind. Graphics created by researcher within InnoColor Edge, 2025.

The user application was initially developed on Ubuntu. It was then deployed on Windows and iOS separately, the latter developed through XCode developer tools, and with the same functions of the Windows application.

In the augmented-reality interface, the toolbar is the collective set of interface buttons on the

left, and the CVD simulation adjustment sliders on the bottom. Within the toolbar, users should select the recoloring algorithm (including Dalt-NET and InnoColor Edge), the direction for CVD simulation (protan, deutan, or tritan), and colors to highlight (between marking out red, green, and blue pixels in the image to aid CVD users, based on HSV values). This is shown in the left image in Fig. 7. The slider at the bottom adjusts the severity level of CVD type. In the order that they are mentioned, the three image transformations are applied to correct the given image, then simulate it for colorblind individuals, and highlight colors for the colorblind individual. Each step can be turned off, creating an adaptive workflow for different users. These three functions comprise the key function of the toolbar, implementing recoloring algorithms including DDPA from Section 3.2 onto the application, providing both CVD users and normal color-vision testers to interface with the InnoColor Edge system.

Specifically geared towards real-time visual aid and testing, the augmented-reality interface contains the camera feed of the iPhone or computer is shown, initially without any modifications. As the user selects options with the toolbar, the camera feed is updated within real-time with the given image processing options. The advantage of this integration of the toolbar is in its flexibility for real-time usage; for example, a protanopic (fully red-blind) user can select a scheme that uses the InnoColor algorithm while highlighting red shades; a color-normal observer can simulate the vision of a deuteranomalous (green-deficient, not fully green-blind) user by selecting the deutan direction, while adjusting the bottom slider to halfway; both colorblind and color-normal observers are recommended to switch between recoloring algorithms to determine which best suits their visual comfort.

In the user-custom data generation interface, both CVD users and color-normal testers alike can process the image series captured within the augmented-reality interface. The interface displays the input and output image side-by-side, where the input image is the raw image captured from the capture, while the output image is from the processed output. Here, the toolbar is available to process each input image and create the processed output image. Each individual image can be saved under different configurations of the recoloring algorithm applied by the toolbar. Finally, the dataset can be exported with each image input-output pair, along with metadata containing the configuration info, allowing datasets to be imported and reproduced when opening the application on another device.

In addition, the TCS3476 RGB ambient light sensor data is displayed on the top right of the augmented-reality interface. The color data point is visualized within the YUV gamut.

Finally, both the iPhone and PC application supports remote control through the Xbox. Through bluetooth integration, the Xbox controller is able to fully navigate through the application, including selecting options within the toolbar, continually adjusting the slider, switching between the two interfaces, and editing data. This functionality is beneficial when the user is testing the iPhone application wearing the InnoColor Edge device, when users are not able to physically interact with their phone screen.

## 4 Results and Discussion

### 4.1 Experimental Details

In this study, the framework was implemented and the model was trained on a server with two Intel(R) Xeon(R) Gold 6248R CPU @ 3.00GHz, 1-TB memory, and eight GeForce RTX 3090 GPUs.

For the teacher model, before the input image was fed into the neural network, it was resized to  $256 \times 256$ , and the patch size was set to 16. For the training phase, the Adam optimizer was adopted, with the hyperparameters set as  $lr = 0.0002$ ,  $\beta_1 = 0.5$ , and  $\beta_2 = 0.999$ . Training was performed for 100 epochs on a single GPU accelerator. For DDPA’s convolutional network, the batch-size was set to 256; training was performed for 200 epochs on four GPU accelerators. The downsampling size,  $M$  was set to 256, while the LUT dimension  $L$  is 17. Other hyperparameters were consistent with those of CVD-Swin.

During quantitative analysis and ablation studies, all 51196 training images were used to determine metrics.

### 4.2 Quantitative Analysis

The performance of the algorithm is shown in Table 1, including the metrics described in 3.2.4; this serves as a comparison of the efficacy of DDPA trained within the InnoColor Edge framework compared to previous algorithms in terms of quantitative measures over a large span of image data. Huang et al.’s hue-rotation (HR) is a lightweight algorithm that preserves naturalness by remapping hue values based on localized histogram equations [12]. However, it does not utilize learning-based methodology, which prevents contextual feature extraction necessary to adapt to the color conditions of natural images. Dalt-NET fuses a series of learned LUTs to provide an adaptive and efficient transformation for recoloring [20]. This method maintains a color consistent transformation, but the architecture is limited in learning power due to the incorporation of RGB values without either differentiation between chromaticity and lightness, or a more comprehensive residual network. The previous InnoColor determines the foreground region of an image through saliency-based segmentation, and applies separate transformations between the foreground and background to maximize the accuracy of salient region detection [27]. Thus, it is able to ensure that a foreground region is detected and visually emphasized. The issue lies in naturalness preservation, in which there is no established mechanism to ensure naturalness between the foreground and background regions while also maintaining the contrast maximization scheme; the transformation is not globally uniform. Additionally, the algorithm relies on segmentation, which is computationally expensive and not robust for processing complex image content. As the recoloring algorithm of the proposed method InnoColor Edge, DDPA achieves promising results in contrast indicators. Of listed comparative loss metrics GC and LC, DDPA results in the lowest value, respectively 0.159 and 0.0178. Recolored outputs from DDPA also demonstrate high contrast within the image itself, as shown in columns GCD, TCC, AGN, where DDPA achieves the highest objective value, respectively 0.506. On average, DDPA also retains the highest SSIM value of 0.873, performing above Huang’s hue

Table 2: Performance comparison of algorithms across contrast, naturalness, and efficiency metrics. Arrows indicate whether higher ( $\uparrow$ ) or lower ( $\downarrow$ ) values are the performance objective. Note that  $CD_0$  and  $CD_1$  represent chromatic difference with luminosity coefficient  $\lambda = 0, 1$  respectively.

| Model             | Contrast        |                 |                |                |                | Naturalness     |                     |                     | Efficiency        |                   |                     |
|-------------------|-----------------|-----------------|----------------|----------------|----------------|-----------------|---------------------|---------------------|-------------------|-------------------|---------------------|
|                   | GC $\downarrow$ | LC $\downarrow$ | GCD $\uparrow$ | TCC $\uparrow$ | AGN $\uparrow$ | SSIM $\uparrow$ | $CD_0$ $\downarrow$ | $CD_1$ $\downarrow$ | Time $\downarrow$ | MACs $\downarrow$ | Params $\downarrow$ |
| Hue Rotation [12] | 0.174           | 0.0189          | 0.469          | 0.810          | 0.208          | 0.850           | 0.0872              | 0.0958              | –                 | –                 | –                   |
| Dalt-NET [20]     | 0.203           | 0.0210          | 0.460          | 0.790          | 0.200          | 0.825           | 0.0926              | 0.1064              | 13.8ms            | 0.311G            | 0.434M              |
| InnoColor [27]    | 0.348           | 0.0346          | 0.416          | 0.720          | 0.237          | 0.695           | 0.2351              | 0.2486              | 67.6ms            | 86.5G             | 217M                |
| DDPA              | 0.159           | 0.0178          | 0.506          | 0.887          | 0.254          | 0.873           | 0.1426              | 0.1489              | 6.43ms            | 0.199G            | 1.85M               |

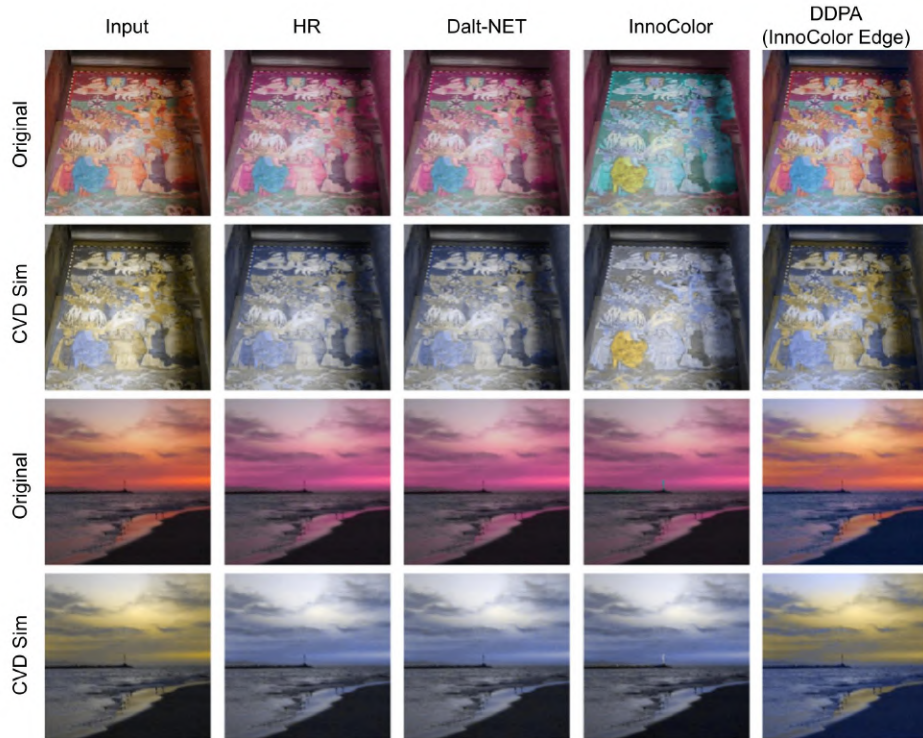


Figure 8: DDPA demonstrates significant improvement in enhancing boundaries of color regions that CVD viewers are likely to confuse, and bring highlight to salient regions that may appear diminished for CVD viewers. Created by researcher with Matplotlib, 2025.

rotation-based algorithm. However, DDPA demonstrates a need for improvement in terms of chromatic difference, especially when comparing to Dalt-NET. In terms of efficiency, DDPA is the fastest for the given image size, processing an image at 6.43 ms. It requires a lower number of calculations, at 0.199G, but retains a higher number of network parameters for LUT creation due to use of large convolutional networks, at 1.85M.

### 4.3 Subjective Evaluation

As shown in Fig. 8, which depicts a natural scene, DDPA improves the local contrast, such as between the people and the ground, while ensuring naturalness. When the original image does not need to be significantly recolored, DDPA maintains a high degree of naturalness. To demonstrate, note that in the carpet of the first row, the shades are primarily light orange or red, which are difficult to distinguish for protanopic observers, as shown in the simulation picture. After recoloring by DDPA, the non-red hues in both the foreground scene and background (which appear as blue) are accentuated, improving the distinguishability of both the foreground from the background and the color features in the foreground itself. In the second example, there are few shades in the scene for normal observers. However, for protanopic observers, the red hue is lost from the clouds, muddying the picture. The result by Huang’s algorithm shifted the red shades to blue, which could be applied for certain cases, but is unsuitable for a natural scene, making the image appear altered. The image generated from the previous InnoColor algorithm significantly recolors the object in the distance but does not similarly recolor the breakwater deck. DDPA maintains the original appearance of the sky, but recolors the gray hues to blue, which enhances the contrast between the ocean color and the distant orange. However, further improvements could slightly reduce the blue hue to maintain more naturalness.

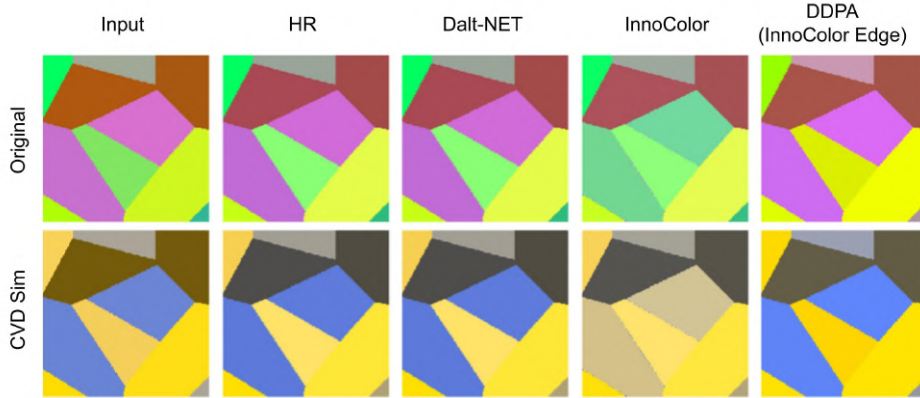


Figure 9: For artificial images, the algorithm demonstrates an ability to distinguish different color clusters and present a potential solution to CVD viewing difficulties in digital GUI tasks. Created by researcher with Matplotlib, 2025.

DDPA also demonstrates potential for recoloring artificial images, often containing bright color pairs that may not be common in natural image datasets. This is demonstrated in Fig. 9. In the first example, adjacent color regions can generally be differentiated, such as the two shades that despite appearing yellow for protanopic observers are clearly distinguishable. Huang’s algorithm causes the two regions to appear closer in color and more difficult to distinguish. The previous InnoColor attempts to locate the salient region of the image, however, this should not be applied for artificial images, as without semantic context, significant color changes may be unwanted. The result is that the purple shades are selected and shifted to an arbitrary color. DDPA mostly preserves the existing image, though it seems to increase the saturation of the purple and yellow regions.

#### 4.4 Performance Testing within Prototype

While Section 4.2 evaluates the performance of DDPA through its ability to converge to optimal values in quantitative metrics, and Section 4.3 analyzes the physical changes conducted by the algorithm onto images, additional testing is needed to both determine the efficacy beyond the algorithm as an integrated hardware-software system and the human-perceived improvements by DDPA.

In previous research, two visual performance criteria are frequently utilized when evaluating recoloring algorithms. These include color naturalness and contrast [5]. Additionally, per the studies of Tanuwidjaja et al., colorblind individuals face discomfort and uncertainty in daily activities, often desiring a sense of reassurance when making decisions. Thus, as an additional criteria of practical usage, saliency is also evaluated [28]. Saliency measures the determinability, and speed of determining, of regions in the image that would constitute the foreground, based on the content of the image. A high level of saliency in recoloring algorithms allows CVD users to quickly determine the primary content of the image, helping to alleviate visual fatigue from prolonged exposure to media that is not adapted to accommodate CVD vision. Thus, a well-designed algorithm suited for real-life use should maintain high performance values across all three criteria.

An experiment is conducted to collect data regarding performance criteria from human trial, involving testing of the recoloring algorithm within the application. A sample of 60 photos were taken within the augmented-reality interface (see Fig. 7). Within the user data interface, each photo was recolored according to one of five algorithms, and then simulated with deuteranopia. Then, a score from 1 to 5 was assigned according to each of the three performance criteria, permitting steps of 0.5.

The experiment was designed with a test-blind procedure: the application was modified to ensure that the algorithm name is not shown during testing, and a program was designed to randomize the order in which an algorithm was applied to a given test image while tracking the reported score.

Human evaluation was conducted across the sample image set for five recoloring algorithms: the previous InnoColor, DDPA, Dalt-NET, hue rotation (HR), and CVD-Swin, the teacher model used to train DDPA. For each criteria, a Wilcoxon signed-rank test is used to determine the advantage of DDPA compared to the other image models (including the baseline input). The test is one-tailed, testing the hypothesis that DDPA exceeds the alternative model. A significant advantage shown by DDPA ( $p < 0.05$ ) is indicated by \*, while a highly significant advantage ( $p < 0.001$ ) is indicated by \*\*. In terms of saliency, DDPA has an extremely significant advantage compared to the input ( $p = 0.0000025$ ), Dalt-NET ( $p = 0.000044$ ), and HR ( $p = 0.0000011$ ), however does not improve relative to the previous InnoColor model or DDPA’s teacher model CVD-Swin. InnoColor conducts binary segmentation on the input image to extract and modify the foreground region, explaining the high saliency. In terms of contrast, DDPA has similar results, receiving higher scores than the input ( $p = 0.000011$ ), Dalt-NET ( $p = 0.000069$ ), and HR ( $p = 0.000018$ ) but not InnoColor or CVD-Swin. Finally, in terms of naturalness, DDPA demonstrates no advantage over Dalt-NET, HR, or CVD-Swin, but receives significantly higher scores than the input ( $p = 0.026$ ) and an extremely

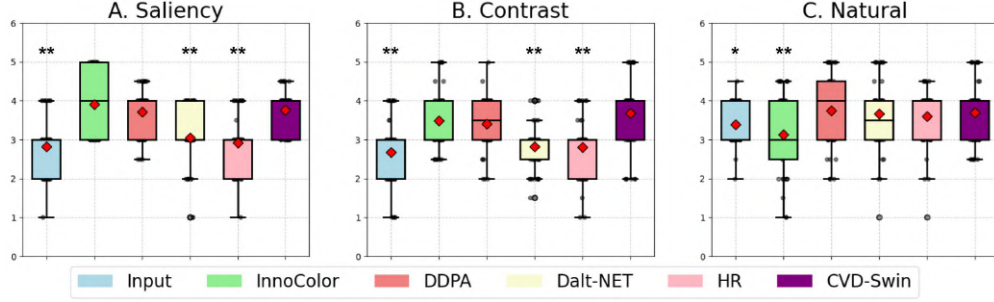


Figure 10: Comparison of Model Performance across performance criteria, based on blind-testing through the wearable device. Boxes are quartiles, while diamonds are the average score.

significant advantage over InnoColor ( $p = 0.00057$ ). A higher naturalness score than the input can be explained due to the lack of clarity and object visibility in the image; however, this suggests that the testing procedure should also be conducted with colorblind individuals as subjects. Throughout all three criterion, DDPA’s performance matches that of the teacher model, signifying the learning capability of its architecture developed in Section 3.2.1.

#### 4.5 Ablation Studies

The ablation study in this study analyzes the impact of three components, including the loss function and feature extraction network.

For the teacher model, CVD-Swin, the SSIM term is removed from the loss function, resulting in contrast-only loss when generating student data [7]. As shown in Figure 11, after training from the contrast-only student data, DDPA enhances the contrast for CVD observers but significantly impairs the realistic qualities of the image, and its conformity to both CVD and normal observer’s perception in real life.

In terms of the student model, DDPA’s training, the impact of TV and MN regularization functions are analyzed. The respective loss coefficients are set to  $\lambda_{TV} = 0$  and  $\lambda_{MN} = 0$  for the DDPA loss function in Eq. 8. As a result, DDPA presents abnormal colors during the interpolation of the 2D-LUT, which hinders image fidelity as shown in Fig. 11. For example, in the left image, the border of the plane is colored distinctly, but is unnoticed by the human eye. However, without LUT regularization, the model will remap the border colors distinctly, creating such unsuitable colors. This demonstrates an issue in convergence, where local LUT values are not resolved.

Next, the impact of the specific feature extraction network is analyzed. This experiment tests the performance of standard convolutional networks Resnet18, MobileNetV3, SqueezeNet, and ShuffleNet [29]. As shown in Table 3, ShuffleNet demonstrates the best results in both contrast and naturalness, with performance close to that of MobileNetV3 for CD, but tending to have better performance in every other quality metric. Resnet18 performs similar to ShuffleNet in every quality metric, except for CD; however, it demands a significantly higher number of MACs and network parameters. MobileNetV3 also presents similar metrics in image contrast and naturalness, but not for comparative loss LC. Specifically, SqueezeNet underperforms compared to the other three models,

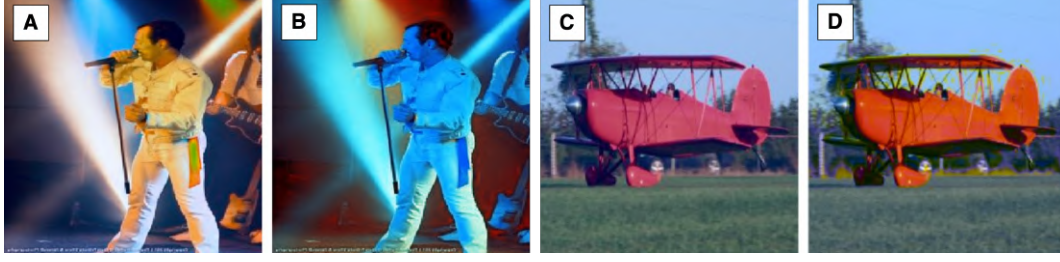


Figure 11: A) With SSIM loss. B) Without SSIM loss. C) With TV and MN. D) Without TV or MN. Created by researcher with Python, 2025.

Table 3: Performance Metrics by Mapping Network (NET) Type. Created by researcher within Overleaf, 2025.

| Model       | Contrast |        |       |       |       | Naturalness |                   |                   | Efficiency |        |         |
|-------------|----------|--------|-------|-------|-------|-------------|-------------------|-------------------|------------|--------|---------|
|             | GC ↓     | LC ↓   | GCD ↑ | TCC ↑ | LAC ↑ | SSIM ↑      | CD <sub>0</sub> ↓ | CD <sub>1</sub> ↓ | Time ↓     | MACs ↓ | Param ↓ |
| Resnet18    | 0.175    | 0.0191 | 0.503 | 0.878 | 0.231 | 0.832       | 0.181             | 0.187             | 4.76ms     | 2.38G  | 11.5M   |
| MobileNetV3 | 0.176    | 0.0259 | 0.504 | 0.876 | 0.227 | 0.853       | 0.144             | 0.148             | 9.91ms     | 0.304G | 4.94M   |
| SqueezeNet  | 0.503    | 0.0437 | 0.329 | 0.581 | 0.158 | 0.398       | 0.650             | 0.674             | 3.82ms     | 0.414G | 1.01M   |
| ShuffleNet  | 0.159    | 0.0178 | 0.506 | 0.887 | 0.254 | 0.873       | 0.143             | 0.149             | 6.43ms     | 0.199G | 1.85M   |

scoring high in loss criteria such as GC and LC, having low image contrast, and failing to maintain naturalness. Regarding time efficiency, all models are similar in magnitude, with SqueezeNet performing the fastest at 3.82ms, and MobileNet being the slowest at 9.91ms. Overall, ShuffleNet maintains an overall advantage in both quality and efficiency.

## 4.6 Protocol for Clinical Applications

Continued outreach and system design in new directions will expand the scope of InnoColor Edge’s potential impact. Initial recruiting will be conducted through InnoColor’s website. Fig. 12 describes the primary experiment protocol for recruited volunteers. Both colorblind and non-colorblind groups are involved. While colorblind participants are blocked based on their condition, the experiment will assign non-colorblind participants into the four colorblind categories. During testing, non-colorblind participants will view media under the simulated colorblindness of assigned types. The focus of this experiment is as follows. First, participants use DDPA to recall the distinguishability of digital images, which are comprised of artificial color palettes, Ishihara test plates, and natural images. Next, users will test the dataset design interface, and feedback will be given, serving as the basis for future iterations of the data pipeline design. Participants are then asked to test the efficacy of the InnoColor Edge system in daily activities through use of the augmented-reality interface within the wearable device. Evaluation will be based on the three criteria established in Section 4.4, and additionally, participant’s measures of visual latency and device comfort. Finally, an interview will be conducted between the researcher and colorblind participants regarding the impact of the condition on daily life. The clinical protocol serves as both a plan for validating the InnoColor Edge system and providing insights into continued development of the research and data pipeline.

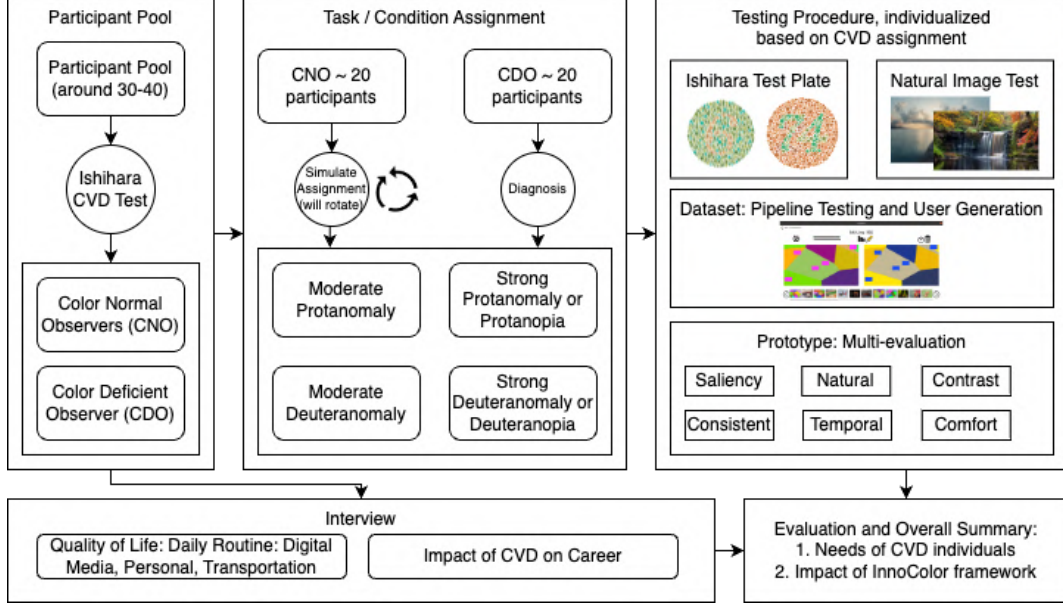


Figure 12: Proposed clinical trial protocol, including experimental block design, unit testing of DDPA, and integration testing of the system. Created by researcher with draw.io, 2025.

#### 4.7 Significance and Limitations of This study

InnoColor Edge’s distillation-based framework enables high-quality recoloring for CVD individuals to be performed within edge-end computational constraints for the first time. Within this framework, through the design of the student model, DDPA, InnoColor Edge improves in contrast metrics such as global chromatic diversity (GCD) by 7.89%, total color contrast (TCC) by 9.51%, average gradient norm (AGN) by 7.17%; structural similarity (SSIM) is improved from the current naturalness-optimized model by 2.71%. DDPA increases computational efficiency, achieving a speed 215% of the next fastest deep-learning algorithm. Finally, it also enables a 36% reduction in MACs. Meanwhile, statistical analysis of human evaluation using the Wilcoxon signed-rank test reveals that DDPA has an advantage in the selected saliency, contrast, and naturalness criteria by extremely significant values, compared to prior models, while maintaining a similar performance of that of CVD-Swin, the teacher model. Additionally, the distillation-based methodology sets a precedent in developing accessible, customizable solutions for visual aid within a user-data pipeline. This extends applicability of the current field to the 95% of colorblind cases (mainly anomalous trichromacy) that are excluded from previous research. Including dichromatic and anomalous trichromatic CVD cases, almost 300 million individuals globally can benefit from utilizing the InnoColor Edge wearable device in multiple aspects of daily life, such as education, professional work, household activities, and entertainment [2, 3]. In fact, InnoColor Edge’s data pipeline is not only limited to recoloring for CVD individuals. The user application, augmented-reality interface, and distillation framework can intake and learn visual recoloring schemes that are psychophysically optimal for individuals afflicted with astigmatism, presbyopia, and amblyopia, for example.

Limitations can still be addressed in this study. While self-testing was conducted through an

experiment, the effectiveness and comfort of the InnoColor Edge system in daily settings can be explored further through clinical research with a larger sample size. Recruiting can be done to determine whether the system provides similar levels of utility across different individuals, and also to begin collection of data for future design iterations throughout the InnoColor Edge pipeline.

## Conclusion

This study achieves three main objectives: (1) to develop a distillation-based image recoloring framework that improves contrast, naturalness, and computational efficiency, (2) to design and implement a wearable device that enables daily visual enhancement for individuals with color vision deficiency (CVD), and (3) to create a mobile and desktop application that integrates this framework into an augmented-reality interface and supports a customizable data generation pipeline. All three objectives were successfully accomplished through the development of the InnoColor Edge system.

First, the Dense Distillation Perceptual Algorithm, trained via a distillation framework from a Swin transformer, achieved substantial advances in algorithmic performance. It enhanced global color contrast ( $GCD = 0.506$ ,  $TCC = 0.887$ ,  $AGN = 0.254$ ) while maintaining high consistency metrics ( $SSIM = 0.873$ ) and computational efficiency (6.43 ms inference, 0.199 MACs). In terms of integrated human evaluation, statistical analysis determines that DDPA significantly improve results in criterion of perceived saliency, contrast, and naturalness within the InnoColor Edge system. These outcomes corroborate the ability of DDPA to effectively resolve prior challenges of low contrast enhancement, color inconsistency, and high computational cost.

Second, the wearable device was designed and fabricated using cost-efficient 3D printing, incorporating a TCS3472 ambient light sensor and bluetooth microcontroller to enhance real-world color adaptation. This prototype demonstrated feasibility for real-time visual augmentation and provided an accessible, ergonomic platform for both end users and researchers.

Third, the InnoColor Edge user application integrated the framework within a real-time augmented-reality environment and provided a data customization and export interface for iterative model re-training. This allows CVD users and developers to generate individualized datasets, ensuring that future developments of the system can adapt to diverse perceptual profiles covering over 95% of CVD cases.

These achievements demonstrate that InnoColor Edge is a viable, scalable, and user-centered system for CVD compensation and visual accessibility. This study not only validates the technical feasibility of real-time recoloring on wearable devices, but also establishes a foundation for personalized data pipeline adaptation in future work. Continued optimization of the convolutional network within DDPA and expanded clinical testing will further refine the system application’s usability and comfort in daily environments.

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